SAI SANTOSH CHEETHIRALA

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Backend engineer who ships services and proves they hold up under failure. Built a sharded, Raft-replicated key/value store in Rust with a from-scratch LSM engine, a CI-gated evaluation service with a public API, and two deployed full-stack products.

EXPERIENCE

PESU Intelligence Labs — Research Intern, RAG & LLM Reliability **Bengaluru | Jun 2026 – Jul 2026**

No Single Culprit — Causal Attribution Benchmark & Evaluation Harness for Multi-Hop RAG **github.com/santoshcheethiralame-dot/LINEUP**

Python, pytest, GitHub Actions (CI/CD), REST API, Streamlit, vLLM, BM25, PyTorch, HuggingFace Transformers, Conformal Prediction

- Built and shipped a 7.2K-LOC evaluation service — public REST API, pooled leaderboard, and Streamlit explorer — with 135 automated tests gating every change on GitHub Actions CI
- Ran ~100K teacher-forced inferences across open-weight models (7B-120B) served with vLLM against a live BM25 retriever, sweeping three multi-hop datasets and retrieval depths k=20-50
- Designed a non-circular counterfactual oracle producing leak-free ground-truth labels without ever consulting the method under test, replacing an evaluation loop that had been silently circular
- Replaced single-passage attribution with a calibrated set-valued method, lifting top-1 accuracy from 0.35 to 0.64 at 94% conformal coverage; first-author paper submitting to TMLR

PROJECTS

tessera — Distributed, Sharded, Replicated Key/Value Store (Rust, std-only) **github.com/santoshcheethiralame-dot/TESSERA**

Rust (0 external crates), Raft, LSM-tree, Deterministic Simulation Testing, Linearizability Checking, MVCC

- Built a distributed, sharded, replicated key/value store in pure Rust with zero external crates: a from-scratch LSM storage engine (memtable, SSTables, write-ahead log, size-tiered compaction, MVCC) and per-shard Raft replication behind a coordinator and routing client (CockroachDB/TiKV shape, scaled down)
- Implemented snapshots and log compaction, linearizable reads via leader leases, joint-consensus membership changes, and crash-safe persistence (term/vote/log fsync'd, temp-file + atomic rename); benchmarked p50/p99 latency and throughput on real disk over a real TCP cluster
- Caught a real split-brain bug via seed 99 — a duplicated vote reply let a candidate win a sub-majority, producing two leaders in one term — under deterministic simulation testing that replays any failure exactly from its seed

GlassBox — Full-Stack Model Interpretability Platform (Deployed) **numero00-glassbox.hf.space**

Python, FastAPI, React, TypeScript, Docker, REST APIs, PyTorch, TransformerLens, SAELens

- Built and deployed a zero-setup browser platform (FastAPI + React + Docker on Hugging Face Spaces) serving 10 live probes over 124M-2.6B-param transformers behind REST APIs
- Ran a 2.6B model on a free tier by diagnosing the load as host-RAM-bound rather than VRAM-bound and re-architecting with CPU-staged low-memory weight conversion to hold peak under a 12 GB ceiling

BATCH — Consumer Product Safety Scanner (Deployed) **batchscan.vercel.app**

TypeScript, Next.js, Open Food Facts, GS1 Digital Link, FDA/CPSC enforcement APIs, Europe PMC

- Shipped a deployed consumer scanner that scores packaged food 0-100 across three separately-openable sub-scores, integrating Open Food Facts, FDA and CPSC enforcement feeds and Europe PMC — backed by 972 tests that run with no network or API key
- Parsed GS1 Digital Link 2D codes for batch and expiry and recovered additive INS numbers that databases routinely miss from Indian ingredient text, designing the parser to refuse rather than guess on unrecognised identifiers

TECHNICAL SKILLS

Languages: Python, C, Rust, TypeScript, JavaScript, SQL, WGSL

Systems & Backend: Distributed systems (Raft, LSM storage, MVCC), FastAPI, REST APIs, Docker, Linux, WebGPU/WGSL, vLLM

Web & Product: Next.js, React, TypeScript, FastAPI, REST APIs, Vercel, Docker, Postgres (Neon), local-first/offline design

Testing & Reliability: Deterministic simulation testing, randomized fault injection, linearizability checking, regression testing, pytest, CI/CD (GitHub Actions), model evaluation

ML & Research: PyTorch, HuggingFace Transformers, vLLM, RAG, conformal prediction, model evaluation

EDUCATION

PES University — B.Tech, Computer Science Engineering **Bengaluru | Sep 2024 – May 2028 (Expected)**

CGPA 7.95/10 (through Sem 4) — CNR Merit Scholarship (top 25% cohort), Sem 1 — Academic Distinction Award (SGPA 7.75+), Sem 2 & Sem 4

Relevant Coursework: Data Structures & Algorithms, Design & Analysis of Algorithms, Operating Systems, Computer Networks, Machine Learning, Deep Learning, NLP, Linear Algebra, Data Science

LEADERSHIP & ACTIVITIES

Sahay AI — Student AI Club, PES University — Multimedia Domain Head **Oct 2025 – Mar 2026**

- Built a content pipeline and posting schedule for a 50+ member AI club, driving 5-6x growth in content views over three months
- National-Level Air Rifle Competitor - 10m Air Rifle, National School Games (a precision event built on sustained focus)